MATH 10A — STRUCTURE AND CONCEPTS IN MATH I

§6.1

Introduction to Fractions

Fractions, unit fractions, four models, and equivalent fractions

Today's Goal

After completing this section, you should be able to:

- Explain what a fraction is, identify its numerator and denominator, and explain how fractions and integers fit inside the rational numbers
- Read and write unit fractions, and describe any fraction as a number of unit fractions
- Represent a fraction with the set model, the bar model, the area model, and the number line model
- Use a model to explain why two fractions are equivalent

Learning Target 14

LT.14 Comparing Fractions

I can apply multiple strategies for comparing fractions.

What Does a Fraction Mean?

A friend says, "I ate $\frac{3}{4}$ of a pizza." What does $\frac{3}{4}$ mean here?

To understand what a fraction represents, we need to identify three things:

1. What is the whole?
2. How many equal parts is the whole divided into?
3. How many of the equal parts do we have?

Here the whole is one pizza. It is divided into 4 equal parts, and your friend ate 3 of those parts.

Fractions

Definition 6.1.1

A **fraction** is a number that represents part of a whole. It is written in the form $\frac{a}{b}$, where

- a is called the **numerator** and
- b is called the **denominator**

The denominator tells us how many equal parts the whole is divided into and the numerator tells us how many of those parts we have.

Note 6.1.2

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1. Fractions are non-negative numbers.
2. A fraction whose value is equal to 0 means that we have none of the equal parts.
3. A fraction whose value is equal to 1 means that we have all of the equal parts.

Note 6.1.2, Continued

Note 6.1.2, continued

4. A fraction whose value is between 0 and 1 (not including 1) is typically called a **proper fraction**.
5. A fraction whose value is greater than or equal to 1 is typically called an **improper fraction**.

How Many Crayons Did Her Friend Get?

Rosa has 20 crayons. She gave $\frac{1}{4}$ of the crayons to her friend. How many crayons did her friend get?

1. The whole is the 20 crayons Rosa has.
2. The denominator is 4, so the whole is being divided into 4 equal parts.
3. The numerator is 1, so we have 1 of the equal parts.



to her friend

So Rosa's friend got 5 crayons.

Rational Numbers

Definition 6.1.5

A **rational number** is any number that can be written in the form $\frac{m}{n}$, where m and n are both integers and $n \neq 0$.

The set of rational numbers includes all fractions and the opposite of each fraction. It also includes all integers because we could choose $n = 1$.

Is -3 a rational number?

Yes. We can write $-3 = \frac{-3}{1}$, and -3 and 1 are both integers.

Unit Fractions

Definition 6.1.6

A **unit fraction** is a fraction that represents one of the equal parts of the whole. A unit fraction is written as $\frac{1}{b}$, where b is the number of equal parts the whole is divided into.

Unit Fraction	Word Form
$\frac{1}{2}$	one-half
$\frac{1}{3}$	one-third
$\frac{1}{4}$	one-fourth
$\frac{1}{5}$	one-fifth
$\frac{1}{6}$	one-sixth

How many one-fifths make $\frac{4}{5}$?

$\frac{4}{5}$ means that the whole is divided into 5 equal parts and we have 4 of those equal parts. So $\frac{4}{5}$ is 4 one-fifths.

The Set Model

In the set model for fractions, a set of objects is taken to represent the whole and subsets of these objects represent fractions of the whole.

Twenty-four students are in a class. $\frac{3}{8}$ of them walk to school. The rest take the bus. How many students walk to school? What fraction of the students take the bus?

The set of objects is the 24 students. This is the whole.

We can think of $\frac{3}{8}$ as 3 one-eighths. One-eighth of the students can be found by dividing the 24 students into 8 equal groups, giving us 3 students in each group.



The Set Model, Continued

Twenty-four students are in a class. $\frac{3}{8}$ of them walk to school. The rest take the bus. How many students walk to school? What fraction of the students take the bus?



3 one-eighths walk

5 one-eighths take the bus

3 one-eighths gives us $3 \times 3 = 9$ students walk to school. The other 5 one-eighths (or $\frac{5}{8}$) take the bus.

The Bar Model

In the bar model (also known as the strip model) for fractions, we divide a rectangular bar (or strip) into equal parts. Each part represents a unit fraction of the whole depending on how many parts there are in total. Below, we have a bar split into 5 equal parts with 3 parts shaded. If we take the entire bar to represent the whole, then the shaded part represents the fraction $\frac{3}{5}$.



Identify the Whole

Warning 6.1.10

When using the bar model, it is important to remember to identify what represents the whole. Usually, the entire bar represents the whole, but that is not always the case. In particular, we may do this when the fraction we are representing is an improper fraction greater than one whole. When the entire bar does not represent the whole, we should be careful to make this clear.

Represent a Fraction with the Bar Model

Use the bar model to represent $\frac{5}{6}$.

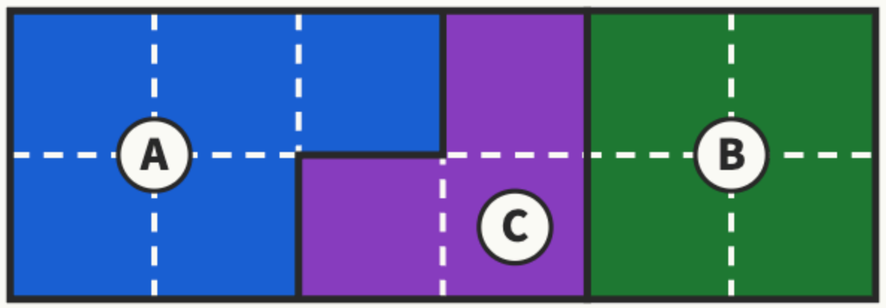
We begin with a bar that represents the whole.



Then we divide that bar into 6 equal parts and shade 5 of these parts.

The Area Model

In the figure below, the large rectangle represents the whole. What fraction is being represented by each of the different colored regions?

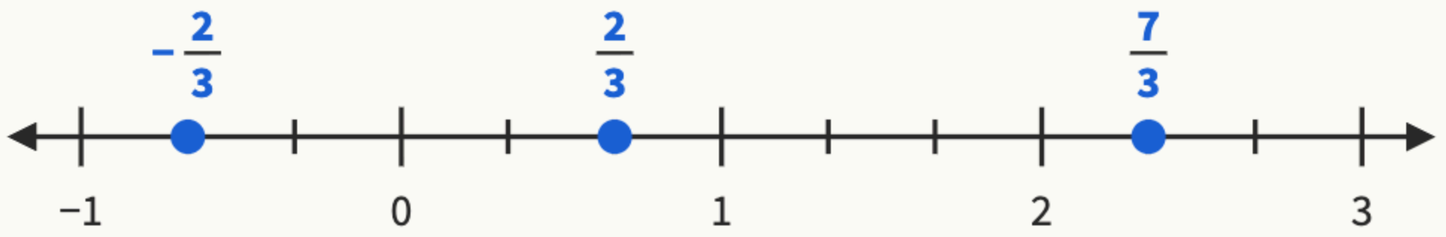


By adding in division lines, we can see that the rectangle is divided into 12 parts of equal area. So each part represents the unit fraction one-twelfth.

Region	Parts	Fraction
blue (A)	5 one-twelfths	$\frac{5}{12}$
green (B)	4 one-twelfths	$\frac{4}{12}$
purple (C)	3 one-twelfths	$\frac{3}{12}$

The Number Line Model

Describe how to locate these numbers on the number line: $\frac{2}{3}$, $-\frac{2}{3}$, $\frac{7}{3}$



Divide the interval from 0 to 1 into 3 subintervals of equal length. $\frac{2}{3}$ is at the end of the second subinterval.

$-\frac{2}{3}$ is the opposite of $\frac{2}{3}$. It is the same distance from 0, but to the left.

Divide the intervals from 1 to 2 and from 2 to 3 the same way. $\frac{7}{3}$ is at the end of the 7th subinterval from 0.

Equivalent Fractions

Definition 6.1.17

Equivalent fractions are fractions that represent the same value or part of the whole.

We can use the different models we learned about to help us identify equivalent fractions.

Equivalent Fractions with the Bar Model

Use the bar model to explain why $\frac{3}{4}$ and $\frac{6}{8}$ are equivalent fractions.

The fraction $\frac{3}{4}$ is represented by dividing the whole bar into four equal parts and shading three of the parts.

The fraction $\frac{6}{8}$ is represented by dividing the whole bar into eight equal parts and shading six of the parts.



By lining up these two bar models, we can see that the shaded regions are the same size. Therefore, the fractions they represent are equivalent because they represent the same part of a whole.

Two Equivalent Fractions

Work with your group. Use two different models to explain why $\frac{3}{5}$ and $\frac{6}{10}$ are equivalent fractions.

You can choose from the four models:

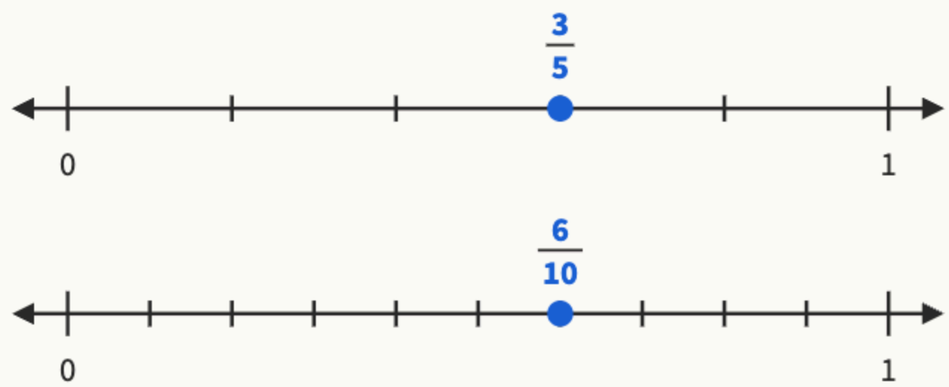
- the set model
- the bar model
- the area model
- the number line model

Two Models, Same Part of the Whole

Bar model. The shaded regions are the same size.



Number line model. The two dots are at the same point.



So $\frac{3}{5}$ and $\frac{6}{10}$ represent the same value, and they are equivalent fractions.

The Multiplication Table

In general, $\frac{a}{b}$ and $\frac{na}{nb}$ are equivalent fractions for any nonzero whole number n.

Here are rows 3 and 5 of the multiplication table. What do you notice?

×	1	2	3	4	5	6	7	8	9	10
3	3	6	9	12	15	18	21	24	27	30
5	5	10	15	20	25	30	35	40	45	50

Reading down each column gives a list of equivalent fractions:
$\frac{3}{5}, \frac{6}{10}, \frac{9}{15}, \frac{12}{20}, \frac{15}{25}, \ldots$ In fact, this would be true if we selected any two rows in the table.